

An imperfection budget for pump lasers in spectrally pure photon-pair sources based on spontaneous parametric down-conversion

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(Dated: July 5, 2026)

Spectrally pure heralded single photons from spontaneous parametric down-conversion (SPDC) are usually engineered assuming an ideal pump laser. We numerically translate three quantities printed on any pump-laser datasheet – center-frequency jitter and drift, relative intensity noise (RIN), and longitudinal-mode structure – into a quantitative budget for the heralded spectral purity. Using a first-order group-velocity-mismatch model of a 10 mm PPKTP-like source (ideal purity 0.810), each imperfection enters as a Monte-Carlo ensemble of pure pump realizations, and the purity is evaluated from the heralded photon’s ensemble density matrix. Center jitter of 26, 61, and 88 GHz rms consumes a 1%, 5%, and 10% relative purity budget, respectively; a re-matched 5 mm crystal roughly doubles these tolerances. Even one additional equal-power longitudinal mode at 0.25 THz spacing exceeds the entire 10% budget. RIN, by contrast, leaves the purity exactly invariant but inflates accidental double-pair emission as $1 + r^2$. Datasheet lines thus map onto two distinct damage channels: spectral (jitter, mode structure) and statistical (RIN).

I. INTRODUCTION

Spontaneous parametric down-conversion (SPDC) remains the workhorse process for generating photon pairs and, by detection of one photon of a pair, heralded single photons; it underpinned the first two-photon interference experiments [1] and the bright polarization-entangled sources that followed [2]. For modern applications the heralded photon must also be spectrally pure. When the joint spectral amplitude (JSA) of the pair carries signal-idler frequency correlations, the heralding measurement leaves its partner in a mixed spectral state, and photons from independent sources become partially distinguishable, degrading quantum interference [3]. The Schmidt decomposition makes this quantitative: the heralded purity is the inverse of the Schmidt number of the JSA [4]. Matching the group velocities of pump and down-converted fields can render the JSA approximately factorable at the source [5], a strategy demonstrated experimentally with pure heralded photons produced without lossy spectral filtering [6].

A mature source-engineering literature has since pushed intrinsic purity toward unity, most prominently by apodizing the nonlinear response through domain engineering of the crystal [7, 8], and spectrally pure heralded sources are now a standard requirement in single-photon-source design [9]. Joint-spectral characterization has also shown that the pump spectrum itself – not only its nominal bandwidth – can limit the purity a real source achieves [10].

In nearly all of this work, however, the pump laser is idealized as a noiseless, transform-limited, single-longitudinal-mode Gaussian. Real pump lasers deviate from this ideal in ways that are already quantified, at no experimental cost, on the manufacturer’s datasheet: center-frequency drift and jitter, relative intensity noise (RIN), and the longitudinal-mode structure of the laser cavity. What has been missing is a translation between

those datasheet lines and the figure of merit a source builder cares about. How many gigahertz of pump jitter cost one percent of heralded purity? Does RIN matter at all?

This paper provides that translation for a concrete, numerically transparent model. We study a collinear SPDC source described by first-order group-velocity-mismatch (GVM) phase matching, with parameters representative of a 10 mm PPKTP crystal pumped at 775 nm and emitting at 1550 nm, with a transform-limited Gaussian pump of 0.164 THz bandwidth (standard deviation); the ideal heralded purity of this reference design is $P = 0.810$. Each pump imperfection is mapped to a Monte-Carlo ensemble of pure pump realizations, and the heralded photon is described by the ensemble-averaged density matrix, from which we compute the mixed-state purity directly. The results form a compact imperfection budget (Table I). Center jitter of 26, 61, and 88 GHz rms consumes a 1%, 5%, and 10% relative purity budget, respectively; re-matching the pump to a 5 mm crystal roughly doubles these tolerances, at the cost of pair rate, which we do not model. Even a single additional longitudinal mode of equal power at 0.25 THz spacing costs 17% of the purity – more than the entire 10% budget; the cost scales with the side mode’s relative power, so a weak, well-suppressed side mode is far less damaging. RIN, by contrast, leaves the heralded spectral purity exactly invariant in this model but inflates accidental double-pair emission by the factor $1 + r^2$ for fractional intensity noise r . Datasheet lines therefore map onto two distinct damage channels: spectral imperfections that degrade purity, and statistical imperfections that degrade pair statistics.

We emphasize the scope: this is a numerical modeling study with first-order (linear) GVM phase matching, quasi-static ensemble noise, and heralded spectral purity (together with the relative double-pair boost) as the figures of merit; absolute pair rate, heralding efficiency, and higher-order dispersion lie outside the model. Section II

defines the JSA, the Schmidt-decomposition purity, and the mixed-state ensemble method. Section III establishes the ideal design baseline (pump-bandwidth and crystal-length optima) and presents the imperfection sweeps behind the budget. Section IV summarizes the budget, develops the spectral-versus-statistical reading of a laser datasheet, and discusses limitations and extensions.

II. MODEL AND METHODS

A. Joint spectral amplitude

We consider collinear, single-spatial-mode SPDC and describe the emitted photon pair by its joint spectral amplitude (JSA). Writing ν_s and ν_i for the angular-frequency detunings of signal and idler from their central design frequencies, the JSA factorizes into an energy-conservation term set by the pump and a momentum-conservation term set by the crystal [3],

$$f(\nu_s, \nu_i) = \alpha(\nu_s + \nu_i) \phi(\nu_s, \nu_i). \quad (1)$$

For a transform-limited Gaussian pump the envelope is $\alpha(\nu) = \exp[-\nu^2/(2\sigma^2)]$, a band along the antidiagonal $\nu_s + \nu_i = 0$; σ is the standard deviation of the amplitude envelope in angular frequency. Bandwidths are quoted throughout as the corresponding ordinary-frequency values; the reference pump has $\sigma/2\pi = 0.164$ THz. For the phase matching we use the standard first-order group-velocity-mismatch (GVM) expansion of the wave-vector mismatch in a crystal of length L ,

$$\phi(\nu_s, \nu_i) = \text{sinc}\left(\frac{\Delta k L}{2}\right), \quad \Delta k = \kappa_s \nu_s + \kappa_i \nu_i, \quad (2)$$

with $\text{sinc}(x) = \sin(x)/x$ and $\kappa_j = 1/v_g^{\text{pump}} - 1/v_g^{(j)}$ the group-velocity mismatch between the pump and photon j . The reference crystal has $L = 10$ mm, $\kappa_s = +0.207$ ns/m, and $\kappa_i = -0.318$ ns/m, representative of PPKTP pumped at 775 nm and producing degenerate pairs at 1550 nm [3, 5, 8]. The ratio κ_s/κ_i sets the tilt of the sinc ridge in the (ν_s, ν_i) plane, and the interplay of this tilt with the pump band determines how separable the JSA can be (Fig. 1). We deliberately retain only this first-order model: no Sellmeier dispersion, no higher-order terms, and no apodization of the nonlinearity [7, 8], so that the imperfection budgets below isolate pump effects at a fixed, transparent phase-matching geometry.

B. Schmidt decomposition

The JSA is evaluated on a square grid of n points per axis over a window of half-width 4–8 THz (chosen per study so that the intensity decays to zero well inside the boundary) and normalized so that $\sum |f|^2 = 1$. The Schmidt decomposition of the discretized JSA is its

singular-value decomposition [4]: with singular values s_k , the Schmidt coefficients are $\lambda_k = s_k^2 / \sum_j s_j^2$, and the heralded-photon spectral purity and Schmidt number are

$$P = \sum_k \lambda_k^2, \quad K = 1/P. \quad (3)$$

$P = 1$ ($K = 1$) corresponds to a fully separable JSA. At the reference point $P = 0.8100$ ($K = 1.2345$).

C. Imperfect pump as a mixed-state ensemble

An imperfect pump is a classical ensemble of pure pump realizations, so the heralded signal photon is described by a density matrix rather than a single Schmidt spectrum. We Monte-Carlo sample $N = 120$ pure realizations ($N = 1000$ for the RIN sweep, whose per-realization cost is lower), compute the discretized JSA matrix A_k for each, and accumulate

$$\rho \propto \sum_k w_k A_k A_k^\dagger, \quad (4)$$

from which the heralded purity follows in a normalization-independent form,

$$P = \frac{\text{Tr}(\rho^2)}{[\text{Tr}(\rho)]^2}. \quad (5)$$

Each imperfection is parameterized directly by laser datasheet statistics and enters only the pump. Per realization we sample: (i) *center jitter/drift* — a Gaussian shift of the pump center frequency with the quoted rms value, treated as a quasi-static shot-to-shot ensemble; (ii) *longitudinal multimode structure* — one comb line drawn from a set of M discrete lines centered symmetrically on the pump, at offsets $(m - (M - 1)/2)$ times the mode spacing for $m = 0, \dots, M - 1$ (half-integer multiples of the spacing when M is even), with normalized line weights (uniform unless stated); the total pump shift is the sum of the comb-line offset and the jitter draw; and (iii) *relative intensity noise* (RIN) — a realization weight $w = \max[0, 1 + \mathcal{N}(0, r^2)]$, where $\mathcal{N}(0, r^2)$ is a zero-mean Gaussian with standard deviation r , the fractional rms intensity noise, each realization's JSA being individually normalized. The 10 mm jitter sweep, which has the largest Monte-Carlo variance, is run with three independent seeds (7, 8, 9) to estimate sampling uncertainty; the remaining Monte-Carlo sweeps use a single seed (7), and for the RIN sweep the heralded purity is exactly independent of the sampled weights (see the validation suite below).

D. Numerical robustness and validation

Ideal-pump sweeps use $n = 512$; Monte-Carlo studies use $n = 256$, which is converged to better than 10^{-4} in P

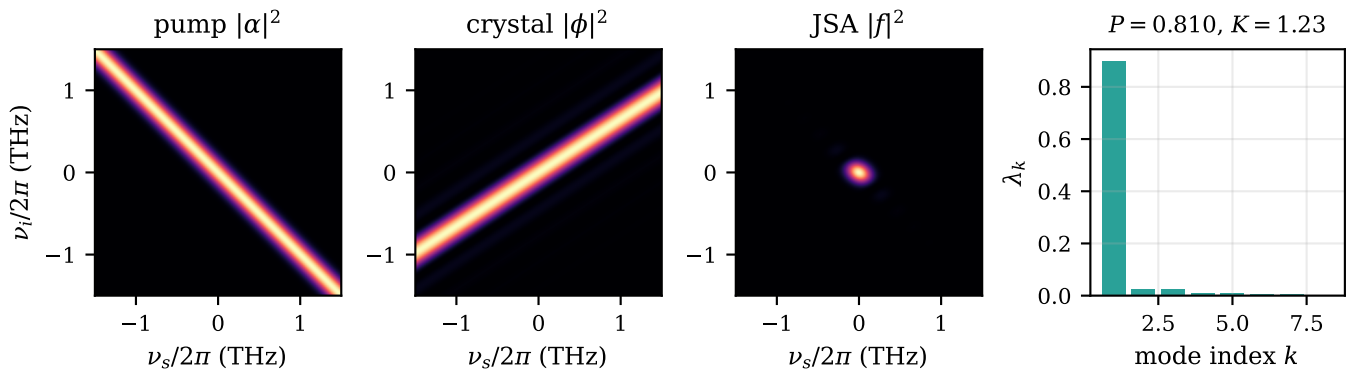


FIG. 1. Anatomy of the JSA at the reference point ($L = 10$ mm, $\sigma/2\pi = 0.164$ THz); axes are detunings $\nu/2\pi$ in THz. Left to right: pump intensity $|\alpha|^2$, a band along the antidiagonal $\nu_s + \nu_i = 0$ (energy conservation); phase-matching intensity $|\phi|^2$, a sinc ridge with side lobes whose tilt is set by κ_s/κ_i (momentum conservation); their product, the joint spectral intensity $|f|^2$, a compact and nearly isotropic central lobe with faint sinc side lobes; and the Schmidt spectrum λ_k of the discretized amplitude, dominated by a single coefficient and giving $P = 0.810$, $K = 1.23$.

against $n = 768$ at the hardest points of the parameter sweeps (0.05 THz bandwidth over a 6 THz half-window; 2 mm crystal over an 8 THz half-window). The RIN sweep uses a coarser $n = 64$ grid: there every realization shares the identical JSA and only the scalar weights vary, so the reported purity invariance is exact at any grid size. At mid-curve jitter (0.25 THz rms) the three seeds give $P = 0.484/0.460/0.513$, a spread of about ± 0.03 , while grid refinement at the same point changes P by only $\sim 2 \times 10^{-5}$: Monte-Carlo noise, not discretization, dominates the error budget. The code ships with a validation suite of nine physics tests, including exact collapse of the zero-imperfection ensemble to the pure-state SVD purity ($P = 0.8100$), exact RIN invariance of the heralded purity (difference below 10^{-9}), reduction of a comb with zero side-line weight to the single-mode result, and monotonic degradation of purity with jitter.

All simulations use the open-source `qsource-sim` package (Python, MIT license); every figure and table in this paper regenerates from a declarative YAML configuration file, one per study.

III. RESULTS

A. Ideal-source baseline

We first establish the ideal-pump baseline that all imperfection budgets refer to. Figure 2(a) shows the heralded purity of Eq. (5) as a function of the pump bandwidth for the 10 mm reference crystal. The curve has the broad interior optimum expected when the pump-envelope and phase-matching correlations of Eq. (1) are balanced [3, 4, 6]: the grid-limited peak is $P_{\max} = 0.8106$ at 0.178 THz on the sweep's grid ($n = 512$, 6 THz frequency half-window), the top is flat, and the canonical reference point $\sigma = 0.164$ THz gives $P = 0.8100$ on the reference grid ($n = 256$, 4 THz half-window) used for

all Monte-Carlo studies below. The 1% tolerance window spans 0.159–0.202 THz, so the bandwidth itself is a forgiving design parameter. The fall-off is nearly log-symmetric about the optimum: equal factors from the peak give nearly equal purities (e.g. $P = 0.663$ and 0.665 a factor of 1.9 below and above it), and the endpoint values— $P = 0.41$ at 0.05 THz against 0.24 at 1.2 THz—differ only because the sweep extends further on the broadband side (factors of 3.6 and 6.7 from the peak, respectively). The residual gap to $P = 1$ is the familiar penalty of an unapodized sinc phase-matching function [7, 8]. Figure 2(b) shows the complementary sweep: at fixed $\sigma = 0.164$ THz the crystal length has its optimum at 10–12 mm ($P = 0.806$ at 10 mm, 0.805 at 12 mm), falling to 0.286 at 2 mm and 0.51 at 30 mm. This is the same matching condition seen from the crystal side. The small offset between the 0.806 quoted here and the 0.8100 reference value above is a frequency-window truncation effect, not discretization: the slowly decaying sinc side lobes make the computed purity drift with the window half-width ($P = 0.8100$, 0.8074, and 0.8061 at the same operating point for the 4, 6, and 8 THz half-windows of the reference grid, the bandwidth sweep, and the length sweep, respectively), so ideal purities are comparable only within a fixed window and are window-independent only at the level $P \approx 0.81$. As a narrowband anchor, a 5 GHz pump yields $P \approx 0.045$ and $K \approx 22$ (Eq. (3)) once the grid resolves the few-GHz pump ridge (which is narrower than the pixel of the sweep grids), recovering the strongly anti-correlated quasi-CW limit.

B. Spectral drift and jitter

Figure 3 shows the heralded purity of the Monte-Carlo mixed state (Eq. (4)) as the pump center frequency jitters with increasing rms amplitude. At zero jitter the ensemble reproduces the ideal value exactly,

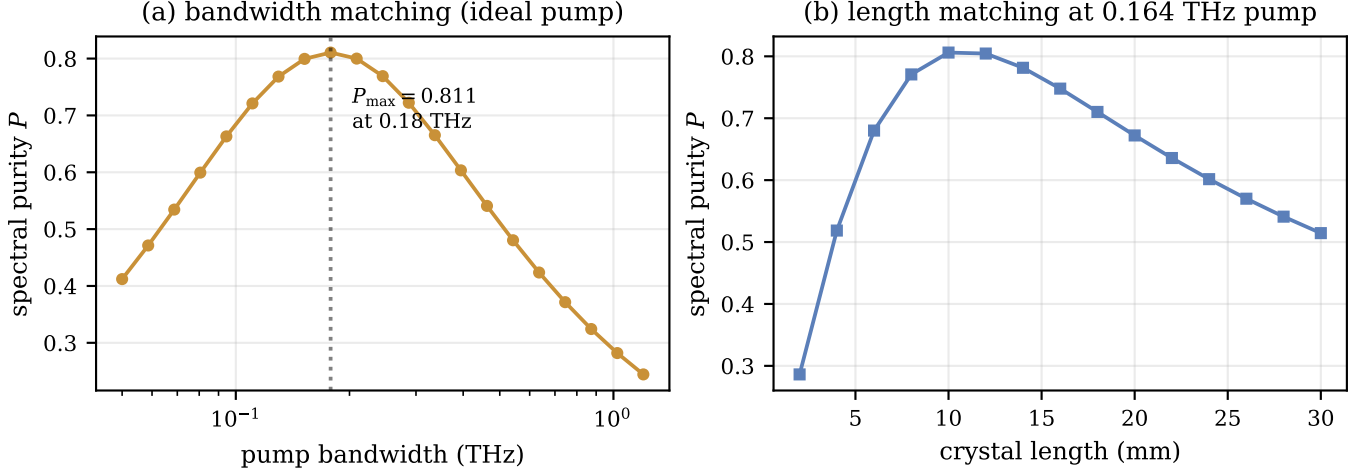


FIG. 2. Ideal-source design space. (a) Heralded purity versus pump bandwidth (log axis) for the 10 mm crystal; the dotted line marks the grid-limited peak $P_{\max} = 0.8106$ at 0.178 THz on a flat-topped optimum. (b) Purity versus crystal length (2–30 mm) at the fixed reference pump bandwidth of 0.164 THz, with the optimum at 10–12 mm.

$P(0) = 0.8100$; the seed-averaged degradation is monotonic and reaches $P = 0.299$ at 0.5 THz rms (individual seeds scatter non-monotonically about this mean, with the worst seed reaching 0.287). The shaded band is the spread of three independent seeds ($N = 120$ realizations each); at 0.25 THz rms the three seeds give $P = 0.484/0.460/0.513$, i.e. a spread of about ± 0.03 that dominates the negligible discretization error ($\sim 2 \times 10^{-5}$ under grid refinement at fixed window). The seed-averaged budgets for the 10 mm crystal are 26/61/88 GHz rms for a 1/5/10% relative purity drop. Re-matching the source with a 5 mm crystal and a 0.328 THz pump (ideal $P = 0.8183$) roughly doubles every budget to 51/121/188 GHz (factors of 1.98–2.12; the 5 mm sweep uses a single seed): the phase-matching bandwidth scales as $1/L$ (Eq. (2)), so a shorter crystal simply presents a wider acceptance to a wandering pump, at the cost of a lower pair rate that our model does not capture.

C. Longitudinal-mode structure

Multimode pump operation is far less forgiving. Figure 4(a) shows the purity ladder for an equal-weight comb at 0.25 THz spacing: $P = 0.810, 0.670, 0.535, 0.451, 0.370, 0.321, 0.297$ for 1–7 lines. A single extra mode at this spacing already costs 17% of the purity—more than the entire 10% budget—so the maximum mode count within any budget is one. Figure 4(b) sweeps the spacing of a fixed 3-mode comb: at 0.02 THz the lines are unresolved by the JSA and $P = 0.807$, essentially the single-mode value, while at 0.6 THz the purity has fallen to 0.31. The spacing budgets are 31/76/111 GHz for 1/5/10%. The practical reading is that a nominally single-frequency pump that occasionally hops or lases on a side mode is a first-order threat to heralded purity, on par with the

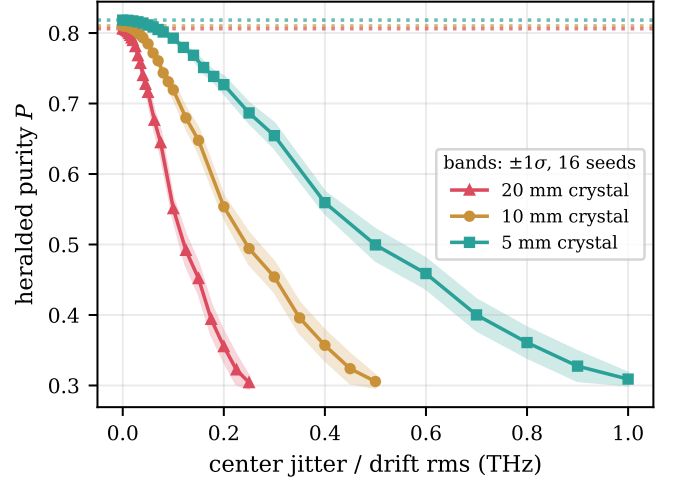


FIG. 3. Heralded purity versus pump center jitter/drift (rms, treated as a quasi-static ensemble) for the 10 mm reference source (circles; shaded band: spread of three Monte-Carlo seeds) and for a 5 mm crystal with the pump re-matched to 0.328 THz (squares). Dotted lines mark the respective ideal purities.

strongest jitter considered above.

D. Intensity noise attacks a different metric

Relative intensity noise behaves qualitatively differently (Fig. 5). Because a pulse-energy fluctuation rescales the ensemble weights w_k in Eq. (4) without changing any normalized spectral amplitude, the heralded purity is exactly invariant: on the $n = 64$ grid used for this study it is flat at 0.809956 for every RIN value from 0 to 30% rms (Fig. 5(a)). What RIN does

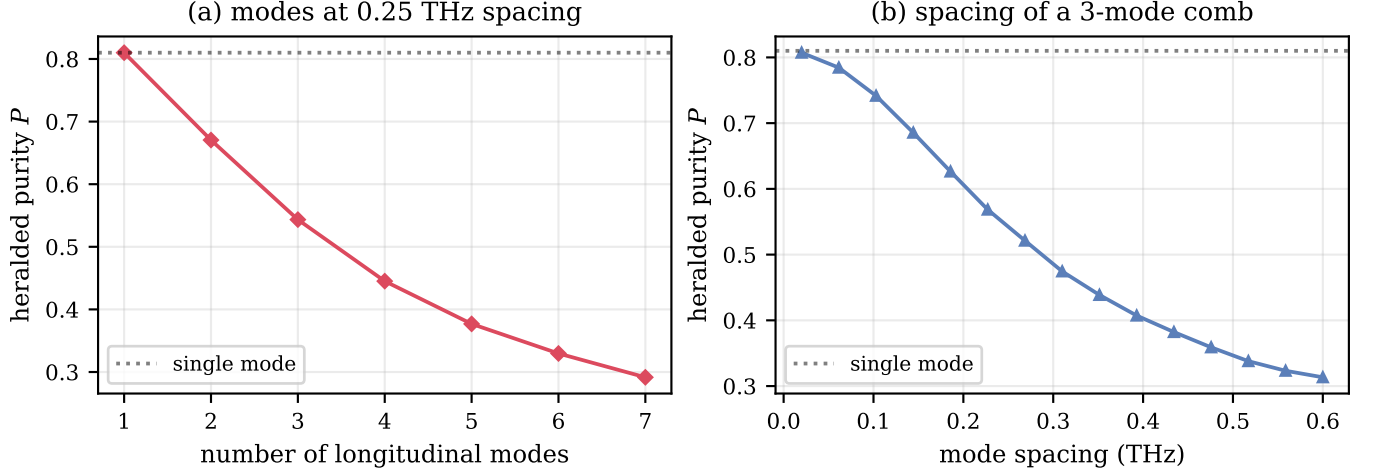


FIG. 4. Longitudinal-mode structure of the pump. (a) Heralded purity versus the number of comb lines at 0.25 THz spacing; the dotted line is the single-mode value. (b) Purity versus the spacing of a 3-mode comb (0.02–0.6 THz); for spacings unresolved by the JSA the single-mode purity is recovered.

damage is the pair statistics relevant for heralding fidelity and accidental double pairs [9, 11]: the double-pair boost $\langle w^2 \rangle / \langle w \rangle^2$ follows the small- r prediction $1 + r^2$, with a measured 1.0221 at 15% RIN against a theoretical 1.0225, and 1.0865 at 30% against 1.09 (Fig. 5(b)). The residual scatter about $1 + r^2$ is finite-sample Monte-Carlo error: the boost estimator has a standard error of ≈ 0.004 at $r = 0.3$ for $N = 10^3$ realizations, and the measured points fall on both sides of the theory (e.g. 1.0560 against 1.0506 at 22.5% RIN), whereas the clipping of the weights $w = \max[0, 1 + \mathcal{N}(0, r^2)]$ at zero is negligible here, shifting the boost by only -8×10^{-5} at $r = 0.30$. Budgeted on the boost, a +1% increase is reached at 10.0% RIN and +5% at 21.7%; +10% is not reached within the 30% sweep. A datasheet must therefore be read with this split in mind: jitter and mode structure are spectral imperfections that degrade purity, while RIN is a statistical imperfection that leaves purity untouched and instead degrades the pair-number statistics.

E. The imperfection budget

Table I collects the results of Figs. 2–5 into a single auto-generated budget. The convention is as follows: each entry is the largest value of the imperfection for which the heralded purity stays within a 1/5/10% *relative* drop of the ideal value of the same source (0.8100 for the 10 mm source, 0.8183 for the re-matched 5 mm source); the pump-bandwidth row is instead the window of bandwidths within the stated drop of the peak, and RIN—which leaves purity exactly invariant—is budgeted on the double-pair boost instead. The hierarchy is unambiguous: longitudinal-mode structure is the least tolerable imperfection, center jitter comes next with budgets in the tens of GHz, and RIN only matters at the ten-percent

level and through a different observable.

IV. DISCUSSION

The central practical message of this study is that a pump-laser datasheet splits into two distinct damage channels. Spectral specifications—center-frequency stability and longitudinal-mode structure—act on the heralded state itself: each realization displaces or replicates the pump band across the phase-matching ridge of Eq. (2), the ensemble average of Eq. (4) mixes inequivalent Schmidt decompositions, and the purity of Eq. (5) falls. Statistical specifications—intensity noise—rescale each realization without changing its spectral shape, so the heralded purity is mathematically invariant (constant to better than 10^{-9} in our validation tests, Sec. II; see also the flat sweep of Sec. III), while the pair-number statistics degrade: the double-pair boost $\langle w^2 \rangle / \langle w \rangle^2 \approx 1 + r^2$ (exact to leading order in r) inflates accidental coincidences and erodes heralding fidelity through exactly the multi-pair channel analyzed in Ref. [11]. Selecting a pump therefore means checking both columns: a spectrally quiet laser with high RIN delivers pure but noisy heralds, while an intensity-stable laser with a wandering center frequency delivers clean statistics on a mixed state.

The budgets also show where a designer has leverage. Jitter tolerance scales approximately as $1/L$: halving the crystal from 10 mm to 5 mm (pump re-matched to 0.328 THz) roughly doubles every budget, from 26/61/88 GHz rms to 51/121/188 GHz rms at the 1%/5%/10% levels (Fig. 3), at the price of a lower pair rate that our model does not capture. The ideal purity itself is unchanged by this trade: the first-order JSA of Eq. (1) is exactly invariant under the rescaling $L \rightarrow L/2$,

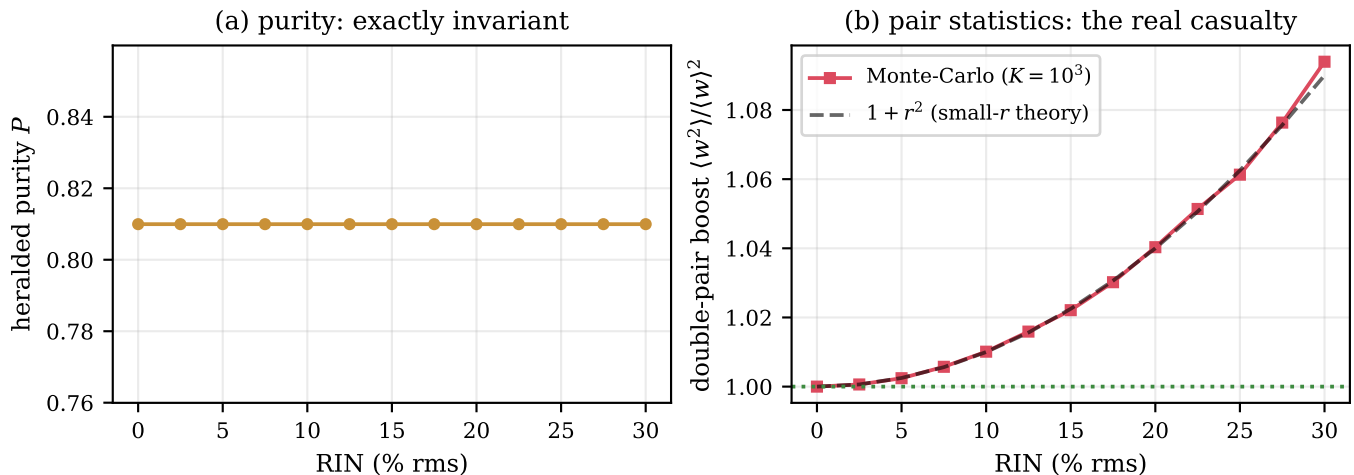


FIG. 5. Relative intensity noise. (a) Heralded purity versus RIN (0–30% rms): exactly invariant, flat at 0.809956 on the $n = 64$ grid. (b) Double-pair boost $\langle w^2 \rangle / \langle w \rangle^2$ from the Monte-Carlo ensemble ($N = 10^3$) against the small- r theory $1 + r^2$ (dashed); the dotted line marks the noiseless value of 1.

Imperfection	unit	budget for relative purity drop		
		1%	5%	10%
Center jitter rms (20 mm)	GHz	13.5 ± 0.2	31.3 ± 0.4	45.3 ± 0.6
Center jitter rms (10 mm)	GHz	27.7 ± 0.3	62.7 ± 0.7	91 ± 1
Center jitter rms (5 mm)	GHz	54.1 ± 0.9	125 ± 1	181 ± 3
Center jitter rms (10 mm, Gaussian PM)	GHz	30 ± 0.3	68.5 ± 0.8	102 ± 1
Mode spacing (3-mode comb)	GHz	29.6 ± 0.1	75.9 ± 0.5	112.1 ± 0.7
Additional side modes (0.25 THz)	count	0	0	0
Pump bandwidth window	THz	0.159–0.202	0.131–0.244	0.114–0.281
RIN (budget on pair boost)	% rms	9.93 ± 0.08	22.3 ± 0.2	beyond sweep

TABLE I. Imperfection budget, auto-generated from the sweeps of Sec. III. Each entry is the largest imperfection keeping the heralded purity within the stated relative drop of the ideal value of the same source; the bandwidth row gives the tolerance window around the purity peak, and RIN is budgeted on the double-pair boost $\langle w^2 \rangle / \langle w \rangle^2$ because the purity is exactly RIN-invariant.

$\sigma \rightarrow 2\sigma$, and the slightly higher value quoted for the 5 mm sweep (0.818 versus 0.810) is a numerical windowing artifact—the fixed frequency window truncates relatively more of the sinc side lobes of the twice-wider 5 mm JSA—not a physical improvement. Bandwidth, by contrast, is the easy specification: the 1% window spans 0.159–0.202 THz, a full width of roughly 26% of the reference bandwidth, and the 10% window spans 0.114–0.281 THz. Center stability is the hard one: the 1% budget of 26 GHz rms corresponds, via $\delta\lambda = \lambda^2 \delta\nu / c$ at $\lambda = 775$ nm, to a wavelength stability of only about 52 pm (≈ 0.05 nm). A pump may exceed its nominal bandwidth by tens of percent with negligible cost—though narrowing below the nominal value is penalized sooner, because the reference bandwidth sits near the lower edge of the tolerance window—but its center must hold to a sub-0.1-nm tolerance.

Our ideal purity of 0.810 is the familiar ceiling of sinc-shaped phase matching, set by the side lobes of the unapodized crystal. Domain engineering and apodization

reshape the phase-matching function toward a Gaussian and raise this ceiling well above 0.81 [7, 8]. Our budgets are *relative* degradations from whatever ceiling the phase matching provides, and the Monte-Carlo ensemble method applies unchanged to engineered crystals: only the sinc factor in Eq. (1) need be replaced. Repeating the sweeps for an apodized source is a direct extension.

Several limitations bound the present results, each pointing to future work. The model uses first-order GVM only: Δk is linearized, with no Sellmeier dispersion or higher-order terms, so the phase-matching ridge is perfectly straight; curvature will matter for broadband pumps and long crystals. We compute heralded purity only; heralding efficiency and absolute pair rate are outside the model, though a complete source assessment must weigh all three jointly [9]. Jitter is treated as a quasi-static shot-to-shot ensemble without explicit slow/fast time structure. An incoherent pump linewidth, beyond the Gaussian coherent bandwidth, is not yet a separate mixed-state channel. Finally, the treatment is

collinear and single-spatial-mode. We note that for two identical, independent heralded sources the Hong-Ou-Mandel visibility equals the single-source purity, $V_{\text{HOM}} = P$ [1, 6], so the budgets reported here translate directly into two-source interference visibility.

V. CONCLUSION

We have converted pump-laser datasheet specifications into a quantitative imperfection budget for the heralded spectral purity of an SPDC source, using a first-order GVM model with Monte-Carlo pump ensembles. The reference source (10 mm crystal, 0.164 THz pump) has ideal purity 0.810. Keeping the purity within 1%/5%/10% of ideal requires center-frequency jitter below 26/61/88 GHz rms; a 5 mm crystal roughly doubles these budgets to 51/121/188 GHz rms at the cost of pair rate. Single-longitudinal-mode operation is mandatory:

a second mode at 0.25 THz spacing alone costs 17% of the purity, and the mode-spacing budgets for a three-line comb are 31/76/111 GHz. Pump bandwidth is forgiving, with a 1% window of 0.159–0.202 THz. Intensity noise leaves the purity exactly invariant but inflates the double-pair boost approximately as $1 + r^2$; the Monte-Carlo budgets place the +1% crossing at 10% RIN and the +5% crossing near 22%. The rule for choosing a pump is simple: spectral specifications damage purity, statistical specifications damage pair statistics, and both columns of the datasheet must be checked.

ACKNOWLEDGMENTS

We thank Prof. Yas Al-Hadithi (King Abdulaziz University) for his guidance and for discussions on laser physics.

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